

✦ USER GUIDE · ACADEMIC EDITION

Cro×word

Puzzle Generator & Assessment Platform

HOW TO USE THE PLATFORM – A GUIDE FOR TEACHERS, STUDENTS & ADMINS

TEACHER

STUDENT

ADMIN

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1 • What is CrossWord?

CrossWord is a web platform for creating, playing, and grading crossword puzzles in a classroom setting. Teachers build puzzles (by hand or with AI from a document), students play them as timed, gamified challenges, and everyone sees live leaderboards and performance analytics.

The three roles at a glance

ROLE	SIGNS IN WITH	CAN DO
Teacher	Teacher ID	Create & publish puzzles, manage a class roster, view analytics
Student	Registration No.	Play published puzzles, view attempts, stats & leaderboards
Admin	Username	Create/remove teachers, view all students platform-wide

How it flows: An Admin creates Teacher accounts → a Teacher uploads their class roster and builds puzzles → the Teacher publishes a puzzle → Students log in, play it, and submit → scores appear instantly on leaderboards and in the Teacher's analytics.

2 · Starting the platform

CrossWord has three pieces: a **PostgreSQL database** (stores everything), a **Django backend** (the API), and a **Next.js frontend** (the website you interact with). Set up the database once, then start the backend and frontend together.

A Set up the database (PostgreSQL)

The backend stores its data in PostgreSQL, so a Postgres server must be installed and running before the backend can start.

```
# install & start PostgreSQL (macOS example)
brew install postgresql@16
brew services start postgresql@16

createdb crossword          # create the database
```

Then open `backend/.env` and fill in your connection details so the backend can reach it:

```
# backend/.env
DB_NAME=crossword
DB_USER=postgres
DB_PASSWORD=your-db-password
DB_HOST=localhost
DB_PORT=5432
```

B Start the backend (API)

```
# from the project folder
cd backend
source venv/bin/activate      # Windows: venv\Scripts\activate
pip install -r requirements.txt # first time only
python manage.py migrate      # builds tables in Postgres (first time)
python manage.py runserver     # serves http://127.0.0.1:8000
```

C Start the frontend (website)

```
# in a second terminal
cd frontend
npm install                # first time only
npm run dev                # serves http://localhost:3000
```

Then open `http://localhost:3000` in your browser. Keep the database, backend, and frontend all running while you use the app.

First-time Admin: after the first `migrate`, create the very first admin account with `python manage.py createsuperuser` in the backend folder. That username/password is your Admin login.

3 • Logging in

The login screen has three tabs — **Teacher**, **Student**, and **Admin**. Each tab is colour-coded so you always know which portal you're entering.

TAB	COLOUR	ENTER
Teacher	Violet	Your Teacher ID + password (given to you by an Admin)
Student	Cyan	Your Registration Number + password (given to you by your Teacher)
Admin	Rose	Your superuser username + password

1. Pick the tab that matches your role.
2. Type your ID and password (use the eye icon to reveal the password if needed).
3. Press **Sign In** — you'll land on your role's dashboard.
4. To leave, use **Sign Out** at the bottom of the sidebar.

4 · For Teachers

Your sidebar has six areas. Here's what each one does.

1 My Puzzles

Your home dashboard. The four tiles at the top summarise **Total Puzzles**, **Assistances** (hints used by students), **Attainment**, and **Average Score**. Below is a table of every puzzle you've made, each with a status of **Draft**, **Published**, or **Archived**, and action buttons:

- **Edit** — open a draft to change its grid or clues.
- **Publish** — make a draft live so students can play it.
- **Archive** — retire a puzzle (hidden from students, kept for records).
- **Analytics** — jump to the performance breakdown for that puzzle.
- **Delete** — remove a draft. (Published puzzles can't be deleted — archive them instead.)

2 Create Puzzle

Two ways to build a puzzle:

- **Manual Entry** — set a title, timer, access window, difficulty, and whether hints are allowed, then *Generate Grid* and add your own clues.
- **AI from Document** — upload a document (PDF/Word) and the platform's AI reads it and drafts clues for you automatically. Review and tweak before publishing.

3 Add Content

Open a puzzle to preview its grid and clue list, make final edits, generate a **printable version** (great for a paper copy or exporting to PDF), and **Publish** when it's ready.

4 Students

Manage your class roster. **Import** students in bulk by uploading a CSV with the format `name, reg_no, password` (one student per row). From the roster you can **reset a student's password** or **remove** a student.

5 Leaderboard

Pick any of your published puzzles to see a ranked table of students by score, completion time, and hints used.

6 Analytics

Per-puzzle insight: average score, completion rate, average time, and the clues students most often get wrong — so you can see where a class is struggling.

STUDENT PORTAL

5 • For Students

Your sidebar is built around playing puzzles and tracking how you're doing.

1 Play Puzzles

A gallery of every puzzle available to you, shown as cards with a difficulty rating. Click a card to open it. Puzzles you've finished are marked **Done** — reopening them lets you review your answers.

Inside a puzzle: type answers into the grid, use **Check Answer** to test a word, reveal a **Hint** if the teacher allowed them (this counts against your score), watch the **timer**, and press **Submit** when you're done. You'll immediately see your score and rank.

SIDEBAR ITEM	WHAT IT SHOWS
My Attempts	Every puzzle you've submitted, with score, time, and rank (🥇🥈🥉 for the top three).
My Stats	Your personal totals and averages across all puzzles.
Leaderboard	How you rank against classmates on each puzzle.
Report	A performance summary of your results.
Re-analyse	Revisit past attempts to review what you got right and wrong.

Note: each puzzle can be submitted once. Play carefully — using hints lowers your final score, so only reveal one when you're stuck.

6 • For Admins

Admins set up and oversee the accounts on the platform. Your sidebar has two areas.

1 Teachers

Create a new teacher by entering a **Teacher ID**, **Name**, and **Password**, then *Create Teacher*. Hand those credentials to the teacher so they can log in. You can also **delete** a teacher from the same list.

2 Students

A read-only, platform-wide view of every student — their name, registration number, the teacher they belong to, and when they were added. Useful for auditing the whole institution at once.

Setup order: create teachers first. Each teacher then imports their own students (Teacher → Students → import CSV), which is why student creation lives in the Teacher portal, not here.

7 • Tips & troubleshooting

SITUATION	WHAT TO DO
Nothing loads / "failed to load"	Make sure the backend is running at <code>127.0.0.1:8000</code> and the frontend at <code>localhost:3000</code> . Both must be running at once.
Backend won't start / DB connection error	PostgreSQL must be running and the <code>crossword</code> database must exist, with matching credentials in <code>backend/.env</code> . Start Postgres, then <code>createdb crossword</code> and re-run <code>migrate</code> .
Student can't see a puzzle	The teacher must Publish it. Drafts and archived puzzles are hidden from students.
Student forgot their password	Their teacher resets it in Teacher → Students → Reset Password.
Teacher forgot their password	An Admin removes and re-creates the teacher account with a new password.
"Puzzle already submitted"	Each puzzle allows one submission per student — this is expected.
Want a paper copy of a puzzle	Teacher → Add Content → printable export, then print or "Save as PDF" from your browser.